

An elementary explicit positivity range for the Li coefficients

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Abstract

Let T be a height to which the Riemann Hypothesis has been rigorously verified by interval arithmetic. We prove by elementary means that the Li coefficients satisfy $\lambda_n \geq 0$ for every integer n with $2 \leq n \leq 2T - O(1)$. For the verified height $T = 3.000175 \cdot 10^{12}$ this gives n up to approximately $6.00035 \cdot 10^{12}$. The proof uses no information about the location of zeros above T : only the non-negativity of the on-line terms, a window bound for the phases, the classical explicit zero-counting estimate, and an elementary bound for the off-line contribution. No hypothesis is assumed. The range obtained is *linear* in T and is therefore superseded by the quadratic ranges that the classical conversion of a verified height yields (see the remark at the end of §9); the interest of the present note is that the argument is elementary and self-contained.

1 Introduction

The Li criterion [1] states that the Riemann Hypothesis (RH) is equivalent to $\lambda_n \geq 0$ for all $n \geq 1$, where

$$\lambda_n = \sum_{\rho} \left[1 - \left(1 - \frac{1}{\rho} \right)^n \right], \quad (1)$$

the sum being over all non-trivial zeros ρ of $\zeta(s)$, counted with multiplicity, in conjugate pairs (so that $\lambda_n \in \mathbb{R}$). Direct numerical verification of $\lambda_n \geq 0$ has been carried out up to $n \leq 10^5$ (see [2, 3] and the references therein). This is not, however, the only kind of range available: converting a verified height into a range, a conversion for which no novelty is claimed here, is classical [6], and, as recorded by [7], already appears in unpublished work of Oesterlé; such ranges grow like a constant times $T^2/(\log T)^2$ and are therefore substantially larger than the linear range proved below. The contribution of this note is accordingly the elementary and self-contained proof, not the size of the range. The purpose of this note is to show that, using only a *verified height* T as input, one obtains an explicit range $n \leq 2T - O(1)$ by elementary estimates, and that no information about the zeros above T is needed beyond a bound on their worst-case contribution.

Throughout, T is a height such that every non-trivial zero with $0 < \gamma \leq T$ has been rigorously verified to lie on the critical line; this is known for $T = 3.000175\,332\,800$ by [8]. We write $N(T) = \#\{\rho : 0 < \gamma \leq T\}$ for the zero-counting function, $F(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + \frac{7}{8}$ and $B_T = \frac{1}{2} \sum_{\gamma > T} \gamma^{-2}$.

2 The phase of an on-line zero

For a zero on the critical line, $\rho = \frac{1}{2} + i\gamma$,

$$1 - \frac{1}{\rho} = \frac{\gamma^2 - \frac{1}{4} + i\gamma}{\gamma^2 + \frac{1}{4}}, \quad \text{so} \quad \left| 1 - \frac{1}{\rho} \right| = 1, \quad (2)$$

and therefore $1 - (1 - 1/\rho)^n = 1 - e^{in\theta_\gamma}$ with

$$\theta_\gamma := \arg\left(1 - \frac{1}{\rho}\right) = \arctan \frac{\gamma}{\gamma^2 - \frac{1}{4}}, \quad \text{hence} \quad \Re[1 - (1 - 1/\rho)^n] = 1 - \cos(n\theta_\gamma) \in [0, 2]. \quad (3)$$

For a zero off the line, write $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$; then

$$\left|1 - \frac{1}{\rho}\right| = e^{c_\rho}, \quad c_\rho = \frac{1}{2} \log\left(1 + \frac{1 - 2\beta}{\beta^2 + \gamma^2}\right). \quad (4)$$

3 Lemma 1: the window bound

Lemma 1. *Let $n \geq 1$ and $\gamma \in [n/2, \min(2n, T)]$. Then*

$$1 - \cos(n\theta_\gamma) \geq C_1(n) := 1 - \cos\left(\frac{1}{2} - \frac{1}{96n^2}\right). \quad (5)$$

Proof. Put $u(\gamma) = \gamma/(\gamma^2 - \frac{1}{4})$ for $\gamma > \frac{1}{2}$, so that $\theta_\gamma = \arctan u(\gamma)$ and

$$u'(\gamma) = \frac{(\gamma^2 - \frac{1}{4}) - 2\gamma^2}{(\gamma^2 - \frac{1}{4})^2} = -\frac{\gamma^2 + \frac{1}{4}}{(\gamma^2 - \frac{1}{4})^2} < 0.$$

Hence $\theta'_\gamma = u'/(1 + u^2) < 0$: θ_γ is strictly decreasing.

For the expansion, put $u = \gamma/(\gamma^2 - \frac{1}{4})$, so that $\theta_\gamma = \arctan u$ and, for $\gamma \geq 2$, $u = \frac{1}{\gamma}(1 - t)^{-1}$ with $t = \frac{1}{4\gamma^2} \in (0, \frac{1}{16}]$. Splitting $(1 - t)^{-1} = 1 + t + t^2(1 - t)^{-1}$ and using $t^2(1 - t)^{-1} \leq \frac{1}{240}$ gives $u = \frac{1}{\gamma} + \frac{1}{4\gamma^3} + \tau$ with $0 \leq \tau \leq \frac{1}{15\gamma^5}$. Since $u \leq \frac{8}{15} < 1$, the alternating series for the arctangent gives $\arctan u = u - \frac{u^3}{3} + \frac{u^5}{5} - R$ with $0 \leq R \leq \frac{u^7}{7}$; expanding and collecting terms therefore yields the *two-sided* bound

$$\left|\theta_\gamma - \left(\frac{1}{\gamma} - \frac{1}{12\gamma^3} + \frac{1}{80\gamma^5}\right)\right| \leq \frac{10}{\gamma^7} \quad (\gamma \geq 2), \quad (6)$$

in which the remainder's sign is controlled, unlike in an asymptotic expansion. Consequently, by monotonicity and $\min(2n, T) \leq 2n$, for $n \geq 15$

$$n\theta_\gamma \geq n\theta_{2n} \geq \frac{1}{2} - \frac{1}{96n^2} + \frac{1}{2560n^4} - \frac{5}{64n^6} \geq \frac{1}{2} - \frac{1}{96n^2},$$

the last step being equivalent to $n^2 \geq 200$; and for $n \geq 7$

$$n\theta_\gamma \leq n\theta_{n/2} \leq 2 - \frac{2}{3n^2} + \frac{2}{5n^4} + \frac{1280}{n^6} < 2 < \pi,$$

the middle inequality being equivalent to $\frac{2}{3}n^4 - \frac{2}{5}n^2 - 1280 \geq 0$. For the remaining finitely many $1 \leq n \leq 14$ the same two inequalities are checked directly on the window, where by monotonicity the extremum is attained at $\gamma = \min(2n, T)$; the values are tabulated in `scripts/PAPERA_lemma1_rigor.py`. Since $1 - \cos x$ is increasing on $[0, \pi]$ and $n\theta_\gamma \in [\frac{1}{2} - \frac{1}{96n^2}, 2]$, the assertion follows. $\square$

Remark 2. Numerically (6) reproduces the exact value of θ_γ to $2.2 \cdot 10^{-10}$ at $\gamma = 10$, to $2.3 \cdot 10^{-17}$ at $\gamma = 100$ and to better than 10^{-18} for $\gamma \geq 10^4$. The quantities $C_1(n)$ equal $0.1223675 \dots$ at $n = 10$ and $0.1224174381 \dots$ for $n \geq 10^4$.

4 Lemma 2: counting

Lemma 3 ([4, Thm. 1], explicit Riemann–von Mangoldt). *For all $T \geq e$,*

$$\left| N(T) - F(T) \right| \leq R(T) := 0.112 \log T + 0.278 \log \log T + 2.510 + \frac{0.2}{T}.$$

Consequently, for $n \geq 2e$,

$$N(T) - N(n/2) \geq [F(T) - R(T)] - [F(n/2) + R(n/2)].$$

Remarks on the three constants and on the last term. The constants 0.112, 0.278, 2.510 are the bound on $|S(T)|$ proved in [4, Thm. 1]; the sequel [5] lowers them to 0.111, 0.275, 2.450, which would only improve the range below. The last term is elementary and we verify it here rather than quoting it. From the exact identity $N(T) = \frac{1}{\pi}\theta(T) + 1 + S(T)$, where $\theta(T) = \arg \Gamma(\frac{1}{4} + \frac{iT}{2}) - \frac{T}{2} \log \pi$ with the continuous branch, one has

$$N(T) - F(T) - S(T) = \frac{r(T)}{\pi}, \quad r(T) := \theta(T) - \left[\frac{T}{2} \log \frac{T}{2\pi} - \frac{T}{2} - \frac{\pi}{8} \right].$$

The explicit Stirling expansion gives $r(T) = \frac{1}{48T} + O(T^{-3})$, so $|r(T)|/\pi \leq 0.2/T$ for $T \geq e$ with a factor 30 to spare; `scripts/PAPERA_trudgian_remainder_check.py` confirms $r(T) \cdot 48T \rightarrow 1$ numerically over the range used. Hence Lemma 3 follows from the quoted bound on $|S(T)|$ alone. $\square$

5 Lemma 3: the off-line contribution

Lemma 4. *Let $\rho = \beta + i\gamma$ with $\beta < \frac{1}{2}$ and $\gamma > T$. Then $\Re[1 - (1 - 1/\rho)^n] \geq -(|1 - 1/\rho|^n - 1) \geq -\frac{n}{2\gamma^2} e^{n/(2\gamma^2)}$. Moreover every zero with $\beta > \frac{1}{2}$ contributes non-negatively.*

Proof. Since $\beta \geq 0$ and $\beta^2 + \gamma^2 \geq \gamma^2$,

$$c_\rho = \frac{1}{2} \log \left(1 + \frac{1 - 2\beta}{\beta^2 + \gamma^2} \right) \leq \frac{1}{2} \log \left(1 + \frac{1}{\gamma^2} \right) \leq \frac{1}{2\gamma^2},$$

using $\log(1 + x) \leq x$. Writing $w = 1 - 1/\rho$ we have $|w| = e^{c_\rho}$ and $\Re[1 - w^n] \geq 1 - |w|^n = -(e^{nc_\rho} - 1) \geq -(nc_\rho)e^{nc_\rho}$ by $e^x - 1 \leq xe^x$ for $x \geq 0$; inserting $c_\rho \leq \frac{1}{2\gamma^2}$ gives the claim. If $\beta > \frac{1}{2}$ then $c_\rho < 0$, so $|w| < 1$ and $\Re[1 - w^n] \geq 1 - |w|^n \geq 0$. $\square$

6 Lemma 4: an explicit bound for the tail constant

Lemma 5. $B_T = \frac{1}{2} \sum_{\gamma > T} \gamma^{-2}$ satisfies, for $T \geq e$,

$$B_T \leq \frac{1}{2} \left[\frac{1}{\pi} \frac{1 + \log \frac{T}{2\pi e}}{T} + \frac{7}{8T^2} + \frac{0.112(1 + 2 \log T)}{2T^2} + \frac{0.278(1 + 2 \log \log T + (\log \log T)^2)}{2T^2} + \frac{2.510}{T^2} + \frac{0.2}{2T^3} \right].$$

Numerically this is $\leq 3.4015 \cdot 10^{-6}$ for $T = 1.13249 \cdot 10^6$ and $\leq 2.8531 \cdot 10^{-12}$ for $T = 3.000175 \cdot 10^{12}$, i.e. between 3.2 and 3.7 times the nominal value $(\log T + 1)/(4\pi T)$.

Proof. By Abel summation, $\sum_{\gamma > T} \gamma^{-2} = \int_T^\infty x^{-2} dN(x) = -N(T)/T^2 + 2 \int_T^\infty N(x) x^{-3} dx$. Bound $N(x) \leq F(x) + R(x)$ by Lemma 3 and drop the (negative) term $-N(T)/T^2$; the six integrals that arise are elementary:

$$2 \int_T^\infty \frac{x}{2\pi} \log \frac{x}{2\pi e} x^{-3} dx = \frac{1 + \log \frac{T}{2\pi e}}{\pi T}, \quad 2 \int_T^\infty \frac{7}{8} x^{-3} dx = \frac{7}{8T^2},$$

$$2 \int_T^\infty 0.112 \log x x^{-3} dx = \frac{0.112(1 + 2 \log T)}{2T^2}, \quad 2 \int_T^\infty \frac{2.510}{x^3} dx = \frac{2.510}{T^2}, \quad 2 \int_T^\infty \frac{0.2}{x^4} dx = \frac{0.2}{2T^3},$$

and $\int_T^\infty \log \log x x^{-3} dx \leq \frac{1+2 \log \log T + (\log \log T)^2}{2T^2}$ for $T \geq e$. $\square$

7 Main theorem

Theorem 6. *Assume that every non-trivial zero with $0 < \gamma \leq T$ lies on the critical line, and that*

$$\#\{\gamma \in [n/2, \min(2n, T)]\} \geq \frac{n B_T^{\text{exp}}}{C_1(n)}, \quad (7)$$

where B_T^{exp} is the explicit bound of Lemma 5 and $C_1(n)$ is as in Lemma 1. Then $\lambda_n \geq 0$. Consequently, with the range $1 \leq n \leq 10^5$ supplied by the direct verifications of [2, 3], one has $\lambda_n \geq 0$ for every integer n with $1 \leq n \leq N_{\max}(T)$, where $N_{\max}(T)$ is the largest n satisfying (7). For $T = 3.000175 \cdot 10^{12}$ this gives $N_{\max}(T) = 2T - O(1) \approx 6.00035 \cdot 10^{12}$.

Proof. Write the sum (1) over the zeros with $0 < \gamma \leq T$ and those with $\gamma > T$. By hypothesis every zero in the first group lies on the critical line, so by (3) its contribution equals $1 - \cos(n\theta_\gamma) \geq 0$; restricting to the window $\gamma \in [n/2, \min(2n, T)]$ and applying Lemma 1 gives

$$\sum_{0 < \gamma \leq T} \left[1 - \left(1 - \frac{1}{\rho} \right)^n \right] \geq C_1(n) \#\{\gamma \in [n/2, \min(2n, T)]\}.$$

For the second group Lemma 4 gives a lower bound $-\sum_{\gamma > T} \frac{n}{2\gamma^2} e^{n/(2\gamma^2)} \geq -n B_T e^{n/(2T^2)}$, using $n < 2T$ and $\gamma > T$ so that $n/(2\gamma^2) < 1/T$; for $n \leq 2T$ the factor $e^{n/(2T^2)} = 1 + O(1/T)$ is absorbed into the explicit constant. Combining,

$$\lambda_n \geq C_1(n) \#\{\gamma \in [n/2, \min(2n, T)]\} - n B_T^{\text{exp}},$$

which is non-negative precisely when (7) holds. The asymptotic claim $N_{\max} = 2T - O(1)$ follows because the count in (7) is $N(T) - N(n/2)$, which by Lemma 3 is $\gg n \log n$ while the right-hand side is $O(\log T)$; the deficit is a constant because near $n = 2T$ the count equals the number of zeros in $[n/2, T]$, of order $(T - n/2) \frac{\log T}{2\pi}$, and the required bound is ≈ 140 for $T = 3 \cdot 10^{12}$, giving $T - n/2 = O(1)$. $\square$

Remark (optimality of the range in T). The ceiling $n \leq 2T$ is not an artefact of the estimate for the off-line zeros, and it cannot be raised by improving that estimate. Two observations make this precise. First, the phase-window argument requires the window $[n/2, \min(2n, T)]$ to lie inside the region where the zeros are known to be on the critical line; for $n > 2T$ the window degenerates to $[n/2, T]$ and is empty once $n/2 > T$, so $n \leq 2T$ is a hard boundary of the method. Second, the two quantities being compared are both linear in n : the on-line count in the window is of order

$(1.5n/2\pi)\log n$, while the off-line cost is at most nB_T^{exp} . The criterion (7) is therefore independent of n up to a factor $\log n$, and the off-line term is never the binding constraint. Consequently, within this method the range is optimal in T , and it can be extended only by verifying the hypothesis to a greater height — a different mechanism, not a sharper estimate, would be needed to go beyond it. The companion note [11] sharpens the treatment of the sum over distant zeros in a different direction, namely for the input inequality of the classical converse theorem, but for the reason just given it does not widen the range here.

8 Numerical verification

- (i) *Calibration.* For $n = 1$ the on-line part of (1) is $2\sum_{\gamma>0}(1 - \cos \theta_\gamma)$; using the zeros of [9] this equals 0.0230938677, against the exact value $\lambda_1 = \frac{1}{2}\gamma_E + 1 - \frac{1}{2}\log 4\pi = 0.0230957089661210338$, a relative agreement of $8 \cdot 10^{-5}$. Including the analytic tail improves the agreement to about 10^{-5} .
- (ii) *Exhaustive verification for every $n \leq 20000$.* Computed with a split of the sum into a directly summed near part and an asymptotically expanded far part (calibrated against direct evaluation to relative error $< 6 \cdot 10^{-8}$), the minimum over all $n \leq 20000$ of $A_T(n)/n$ is $1.1547 \cdot 10^{-2}$, attained at $n = 1$, exceeding the threshold by a factor of $2.2 \cdot 10^4$. The fifteen worst orders are $1, \dots, 15$.
- (iii) *Margins in the window criterion.* With $T = 1.13249 \cdot 10^6$ and the nominal B_T , the ratio of available to required window count is $2.7 \cdot 10^5$ at $n = 10^5$, $3.2 \cdot 10^5$ at $n = 5 \cdot 10^5$, $1.4 \cdot 10^5$ at $n = 10^6$, $1.5 \cdot 10^4$ at $n = 2 \cdot 10^6$ and 1.03 at the endpoint $n = 2\,264\,960 = 2T - 20$.
- (iv) *Required counts under the explicit bound.* The right-hand side of (7) equals 2.78 at $n = 10^5$, 27.8 at $n = 10^6$ and 62.9 at the endpoint, against available counts 234681, 1 182 636 and 19 respectively; this is why the explicit constants of the classical counting bounds are immaterial except at the very endpoint.

9 Scope

The result is a *subset* statement implied by RH, not a proof of RH: the range $n \leq 2T - O(1)$ grows with T but never covers all n . It is unconditional given T , depending only on the classical representation (1), on the rigorously verified zeros below T [8], and on the classical explicit counting bound [4]. No information about the location of the zeros above T is used; only their worst-case contribution is bounded. We do not claim novelty for the conversion of a verified height into a range for λ_n , which is a folklore consequence of $|1 - 1/\rho| = 1$ on the line and $|1 - 1/\rho| > 1$ off it; what is given here is an elementary, explicit, and numerically calibrated form of the resulting range.

Remark (relation to the quadratic ranges; read this before comparing). The range proved here is linear in the verified height. Ranges quadratic in it are also available from the same input by a different route: combining the decomposition of Bombieri and Lagarias [6] with an explicit bound for ζ'/ζ on the line produces ranges of the shape $n \leq cT^2/(\log T)^2$, and a statement of that shape goes back, in unpublished form, to Oesterlé [7]. Such ranges are larger than the one proved here by a factor of order $T/(\log T)^2$, hence by many orders of magnitude at any realistic height. It would therefore be wrong to describe the present result as an improvement over “direct numerical verification up to $n \leq 10^5$ ”: that comparison is between two different kinds of range, and against

the quadratic conversion-type range of the same kind the range here is weaker, not stronger. The value of the note is the elementary route, not the range.

A The uniform variant, and what is provable there

Theorem 6 fixes n and counts zeros in a window; it does not require any estimate uniform in n . It is nevertheless useful to record a stronger variant, because saying exactly what is and is not elementary about it prevents the range from being misread. In the uniform variant one asks, at a verified height T_0 , for

$$\sum_{\gamma \leq T_0} (1 - \cos n\theta_\gamma) \geq n B_{T_0} \quad \text{for every } 1 \leq n \leq T_0^2, \quad (8)$$

with $B_{T_0} = \frac{1}{2} \sum_{\gamma > T_0} \gamma^{-2}$, equivalently $|S_n| \leq N(T_0) - n B_{T_0}$ for $S_n = \sum_{\gamma \leq T_0} e^{in\theta_\gamma}$. This is strictly stronger than what Theorem 6 needs, and the next two statements are the part of it that is elementary.

Proposition 7 (small n , elementary). *Let $T_0 = 1.132490\,658\,714\,411 \cdot 10^6$ and let $\gamma_1 = 14.134\,725 \dots$ be the first zero ordinate. Then (8) holds for every integer n with $1 \leq n \leq \gamma_1$: the inequality is implied by $n \geq B_{T_0} / (\frac{1}{3} \sum_{\gamma \leq T_0} \theta_\gamma^2) = 1.36 \cdot 10^{-4}$, which is immediate.*

Proof. $\theta_\gamma = 2 \arctan \frac{1}{2\gamma} \leq \theta_{\gamma_1} = 0.0707 \dots$, so $n \leq \gamma_1$ gives $n\theta_\gamma \leq 0.99 < 1$: the phases do not wrap. On $[0, 1]$ one has $1 - \cos x \geq x^2/3$ (the function $x \mapsto 1 - \cos x - x^2/3$ is non-negative there), whence $\sum_{\gamma \leq T_0} (1 - \cos n\theta_\gamma) \geq \frac{n^2}{3} \sum_{\gamma \leq T_0} \theta_\gamma^2$. Numerically $\sum_{\gamma \leq T_0} \theta_\gamma^2 = 2.3104 \cdot 10^{-2}$, so the right-hand side is $7.70 \cdot 10^{-3} n^2$, which exceeds $n B_{T_0}$ as soon as $n \geq 1.36 \cdot 10^{-4}$. The margin at $n = 1$ is $\delta = 5.75 \cdot 10^{-9}$. $\square$

Proposition 8 (smooth part, van der Corput). *Write $S_n = S_n^{\text{smooth}} + D_n$, where S_n^{smooth} is formed from the monotone model $\bar{N}^{-1}(k + \frac{1}{2})$ of the zero ordinates, $\bar{N}(t) = \frac{t}{2\pi} \log \frac{t}{2\pi e} + \frac{7}{8}$. Let $\gamma_{\text{res}} = 8.2683 \cdot 10^5$ be the resonance point defined by $|\varphi'| = 1$ for $\varphi(k) = n\theta(\bar{N}^{-1}(k + \frac{1}{2}))$ at $n = T_0^2$, and let M_B be the number of zeros with $\gamma_{\text{res}} < \gamma \leq T_0$. Then $M_B = 581\,491$ and, by the second-derivative test (the phase is monotone with $|\varphi''| = 6.58 \cdot 10^{-7}$ there),*

$$|S_n^{\text{smooth}}| \leq M_B \sqrt{|\varphi''|} + |\varphi''|^{-1/2} = 471.7 + 1232.7 = 1704,$$

while the allowance of (8) at $n = T_0^2$ is $6.546544 \cdot 10^5$: the mean-density model therefore already produces the required cancellation with a factor 384 to spare, the fast region $|\varphi'| \geq 1$ being handled by the first-derivative test with a dyadic shell summation. Consequently every obstruction in (8) sits in the rigid error D_n .

Remark 9 (what does not work, and by how much). Three classes of attempts fail, and it is worth recording the margins so that the range of Theorem 6 is not mistaken for something it is not.

(i) *Rigidity.* Estimating D_n by the second-derivative test on the *actual* zero ordinates splits the sum at γ_{res} : on the fast region the test needs $\max_k |\text{gap}_{k+1} - \text{gap}_k| \leq 0.675$, while the measured maximum is 1.663; the resulting bound exceeds the allowance by a factor 1.5. On the slow region $\gamma < \gamma_{\text{res}}$, containing $1.42 \cdot 10^6$ zeros, the second differences are $1.9976 \cdot 10^{10}$ in size and the test is void altogether. (ii) *The second-derivative test in aggregate form.* Requiring only a bound on the total second-difference mass $\Lambda = \sum_k |\Delta^2 \varphi_k|$ is false: a single full-turn step contributes 4π of mass while leaving the whole sum coherent, missing the truth by a factor 369.5, and a convex parabola

with one-signed second differences defeats the same form by a factor up to 3.35. The standard test needs a sign-definite second derivative, which the sequence of phases here does not have (its second differences are of mixed sign in a split of roughly one half each, dominated by the jumps of the zero gaps). (iii) *A pair-correlation input.* The negative mass of $|S_n|^2$ is spread over all separations rather than concentrated at neighbours, and the double sum is statistically indistinguishable from one built from independent uniform phases; the neighbour term alone is positive, so pair-correlation information controls a single, wrongly signed part of the sum and cannot supply the cancellation. What is missing is not a stronger input but a more *uniform* one: an estimate of the counting-error increments valid simultaneously for all frequencies $\lambda = 2\pi n/(\gamma^2 \log \gamma)$ when $n \leq T_0^2$, of strength $|D_n| = o(N)$. No such statement is available in the literature, and the search for one is a separate line of work [12].

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